NINE CASES IN WHICH THE COTANGENT COMPLEX VANISHES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file cotangent.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The statement's last item is the author's own placeholder entry 'add more here', kept verbatim inside the counted statement span as part of the author text.

Editorial note. The six references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: the retained preamble of this chapter (sources/stacks-cotangent/retained-preamble.tex, lines 1-63), containing the Xy-pic setup, the comment environments, the theorem declarations and the macros including \NL.

Allowed standard prerequisites (editorial).

- The derived diagonal lemma for a surjective flat ring map. Lemma lemma-derived-diagonal, used at proof lines 1852-1855. Gives the vanishing over the tensor square in case (1).
- The bootstrap lemma. Lemma lemma-bootstrap (1804-1809, produced as L44151), used at proof lines 1855-1857. Concludes the vanishing in case (1).
- The cotangent complex commutes with filtered colimits. Lemma lemma-colimit-cotangent-complex (373-377, produced as L44165), used at proof lines 1858-1858. Gives case (6).
- (Strict) henselizations as filtered colimits of etale ring extensions. Algebra, Lemmas algebra-lemma-henselization-different and algebra-lemma-strict-henselization-different, used at proof lines 1859-1862. Gives cases (7) and (8).

Source reference. Original lemma 04GV: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 5846–5846. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 04GW: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 5847–5847. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 08R1: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11079–11079. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Date: stacks-project:cotangent.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 08S9: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11122–11122. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 08R0: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11078–11078. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original definition 092B: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11474–11474. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *The cotangent complex $L_{B/A}$ is zero in each of the following cases:*

1. *$A \to B$ and $B \otimes_A B \to B$ are flat, i.e., $A \to B$ is weakly étale (More on Algebra, Definition 092B),*
2. *$A \to B$ is a flat epimorphism of rings,*
3. *$B = S^{-1}A$ for some multiplicative subset $S \subset A$,*
4. *$A \to B$ is unramified and flat,*
5. *$A \to B$ is étale,*
6. *$A \to B$ is a filtered colimit of ring maps for which the cotangent complex vanishes,*
7. *B is a henselization of a local ring of A,*
8. *B is a strict henselization of a local ring of A, and*
9. *add more here.*

Proof. In case (1) we may apply Lemma 08R0 to the surjective flat ring map $B \otimes_A B \to B$ to conclude that $L_{B/B\otimes_A B} = 0$ and then we use Lemma 08R1 to conclude. The cases (2) – (5) are each special cases of (1). Part (6) follows from Lemma 08S9. Parts (7) and (8) follows from the fact that (strict) henselizations are filtered colimits of étale ring extensions of A, see Algebra, Lemmas 04GV and 04GW. □